Back **Injective and surjective functions**
Practice Assignment • 30 min

Due Jul 6, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your latest score.

Next item

1. A function f is said to be one-to-one (injective) if and only if: 1 / 1 point

- (selected) $a \neq b$ implies $f(a) \neq f(b)$ for all a and b in the domain of f
- $f(a) \neq f(b)$ implies $a \neq b$ for all a and b in the domain of f
- $a \neq b$ implies $f(a) = f(b)$ for all a and b in the domain of f
- $a = b$ implies $f(a) = f(b)$ for all a and b in the domain of f

Nice work

By definition a function is one-to-one if and only if any two different elements of the domain have two different images.

2. A function f is said to be onto (surjective) if and only if: 1 / 1 point

- (selected) the range of f is equal to the co-domain of f
- the range of f is equal to the domain of f
- the range of f is a subset of the domain of f
- the co-domain of f is subset of the range of f

Nice work

A function f is surjective means that every elements in the co-domain of f has at least one pre-image. This means $R_f =$ co-domain of f

3. Let $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = x^2$. Which one the following statements is true about the function f? 1 / 1 point

- (selected) f is not a one-to-one (not injective) function as $f(x) = f(-x) = x^2$, but $x \neq -x$ for all $x \in \mathbb{R}^+$
- f is a one-to-one (injective) function as $f(x) = f(-x) = x^2$, for all $\forall x \in \mathbb{R}$
- f is an onto function as for all $y \in Co - D_f$ there exists $x \in D_f$ such that $f(x) = y$
- f is a one to one function as for all $y \in Co - D_f$ there exists $x \in D_f$ such that $f(x) = y$

Nice work

$f(-2) = f(2) = 4$ hence, f is not a one-to-one function

4. Let $f : \mathbb{Z} \to \mathbb{Z}$ with $f(x) = 2x + 3$. Which one the following statements is true about the function f? 1 / 1 point

- f is a not one-to-one (not injective) function as some elements $\in \mathbb{Z}$ have more than one pre-image.
- f is not a one-to-one (not injective) function as for all $y \in \mathbb{Z}$ there exists $x \in \mathbb{Z}$ such that $f(x) = y$
- (selected) f is a one-to-one (injective) as for all $a, b \in \mathbb{Z}$ where if $a \neq b$ then $f(a) \neq f(b)$
- f is an onto (surjective) function as for all $y \in \mathbb{Z}$ there exists $x \in \mathbb{Z}$ such that $f(x) = y$

Nice work

for all $a.b \in \mathbb{Z}$ we have $a \neq b \implies 2a \neq 2b \implies 2a + 3 \neq 2b + 3 \implies f(a) \neq f(b)$

5. The following graph represents the curve of the function $f : \mathbb{R} \to \mathbb{R}$. 1 / 1 point

Which one of the following statements is true about the function f?

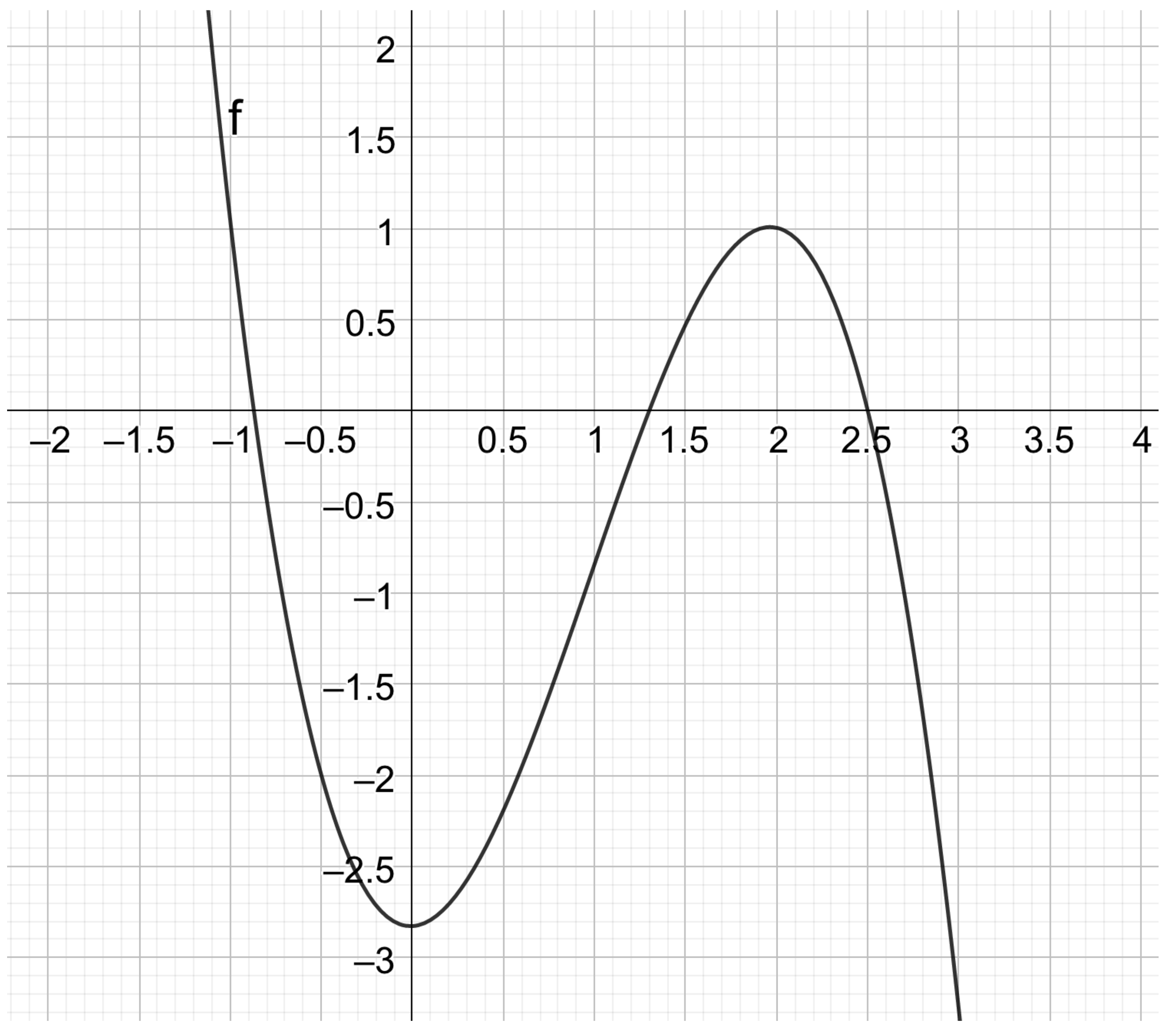

Image description:

The domain of the this function is $\mathbb{R}$ its corresponding curve is decreasing for $x \leq 0$,is increasing for $0 \leq x \leq 2$ and is decreasing for $x \geq 2$.

- (selected) f is surjective but not injective
- f is injective but not surjective
- f is both surjective and injective
- f is neither surjective nor injective

Nice work

f is surjective as every element of $\mathbb{R}$ as a pre-image. However, f isn't injective as the curve shows that at least two elements have the same image. For example, the curve crosses the x-axis at three points, this means that three elements have the image 0.